

Algorithm Analysis & Complexity Theory

Sample Exam: Graph Algorithms

Generated: November 03, 2025

Difficulty Levels: Foundation, Conceptual, Application, Advanced

Question 1 [Difficulty: Advanced]

Prove the correctness of Dijkstra's algorithm for finding shortest paths.

Solution:

Dijkstra's Algorithm Correctness Proof:

Claim: When a vertex v is added to the set S of processed vertices, $d[v]$ equals the shortest path distance from source s to v .

Proof by induction on $|S|$:

Base case: $|S| = 1$

- $S = \{s\}$, $d[s] = 0$
- Shortest path from s to s is 0 ✓

Inductive hypothesis: Assume true for $|S| = k$

Inductive step: Prove for $|S| = k+1$

- Let u be the next vertex added to S
- u has minimum $d[u]$ among all vertices not in S

We need to show: $d[u]$ is the true shortest path distance

Proof by contradiction:

- Suppose $\exists$ shorter path P from s to u
- Let y be the first vertex on P not in S
- Let x be the predecessor of y on P (x is in S)

Since edge weights are non-negative:

- $\text{distance}(s \rightarrow y \rightarrow u) \geq \text{distance}(s \rightarrow y)$
- By IH, $d[x] = \text{distance}(s \rightarrow x)$
- $d[y] \leq d[x] + \text{weight}(x, y) = \text{distance}(s \rightarrow y)$

But u was chosen because $d[u] \leq d[y]$

Therefore: $d[u] \leq d[y] \leq \text{distance}(s \rightarrow y) \leq \text{distance}(s \rightarrow u \text{ via } P)$

This contradicts our assumption that P is shorter

$\therefore d[u]$ is the true shortest path distance ■